Book III

Persistence, Transport, and Observational Transfer

Elements of Nested Fibrational Cosmology

Contents

III.0 Purpose 4
1. Imports and Standing Rules 4
2. Persistent Observables and Transfer Maps 5
3. Persistence Lemmas 6
4. Persistence and Compatibility Propositions 7
5. The Interface-Relative Stability Functional 8
6. Finite Balance and the Unified Coercive-Transport Inequality 9
7. Transfer Coercivity 11
8. Observational-Transfer Objects and POT Preparation 12
9. Governing Theorem of Book III 13
10. Boundary of Book III 15
11. Handoff to Book IV 16
12. Dependency Ledger 17

III.0 Purpose

Book III is the motion and comparison book of the canonical spine.

Book I established that admissible testing produces a coherent observational quotient and that every finite refinement chain stabilizes. Book II established that, under LS-2, stabilized local observable outcomes are determined by bounded collar data and that finite collar alphabets impose hard multiplicity bounds. Book III now asks: how does certified local structure *persist* and *transfer* across admissible refinements and contexts?

This question is essential because Book IV must certify the existence of a universal source object U that all lawful downstream structures factor through. Factorization requires that certified observables from different local regimes can be coherently compared. Coherent comparison requires a transport law. Book III provides that law.

The central theorem of the book is the Unified Coercive-Transport Inequality: a branch-agnostic bound on how the interface-relative stability functional changes under admissible enlargement. This inequality encodes the finite balance between transported bulk stability, internal defect, and boundary defect. Its proof uses only the defect ledger of Book I and the LS-2 boundary law of Book II.

Book III is explicitly branch-agnostic. It does not specialize to Yang–Mills, Navier–Stokes, gravity, or any other specific endpoint program. All such specializations belong in derived branch books. What this book establishes is the upstream source law from which those specializations descend.

1. Imports and Standing Rules

Imports from Book I. Finite configuration spaces Ω, admissible test families $\mathcal{T}_n$, observational equivalence $\sim_n$, quotients Σ_n, stabilized quotient Σ_∞, survivor chains, one-step defect $\delta(C_n)$, defect ledger $\Delta(C_n \rightsquigarrow C_m)$, and the Observational Quotient Theorem.

Imports from Book II. Regions U, collars $\partial_r U$, stabilized collar types, the boundary law map Φ_U under LS-2, realized stabilized outcomes $O(U)$, boundary capacity $\mathrm{Cap}_\partial(U)$, and the Boundary Stabilization Theorem.

Standing Rule 1.1 ([D]). No persistent observable may depend on presentation-level distinctions that were eliminated by the observational quotient of Book I. Persistence is quotient-visible.

Standing Rule 1.2 ([D]). No admissible transfer may destroy the stabilized local structure certified in Book II without recording a transport obstruction. Transfer must respect stabilization.

Standing Rule 1.3 ([D]). Claims about gain, loss, or conservation of observable content must descend from certified finite balance data. Balance before interpretation.

Standing Rule 1.4 ([D]). Observable comparison across contexts has canonical force only through explicitly declared transfer maps.

2. Persistent Observables and Transfer Maps

Definition 2.1 ([D]). A *persistent observable* is a certified observable whose value is preserved or lawfully transformed across the declared class of admissible refinements or comparison contexts. Formally, it is a function $\mathcal{O}$ defined on stabilized quotient classes that satisfies the compatibility condition declared for the relevant class of transfer maps.

Definition 2.2 ([D]). A *transfer map* $f : (U, \Sigma_\infty) \to (U', \Sigma'_\infty)$ is a certified witness-preserving comparison map carrying admissible observable data from one context or refinement stage to another. It must be compatible with observational quotient structure: if $\omega \sim_\infty \omega'$ in the source context, then $f(\omega) \sim'_\infty f(\omega')$ in the target context.

Definition 2.3 ([D]). A *transport class* is an equivalence class of admissible observable data under lawful transfer: two observations are in the same transport class if one is the transfer-image of the other via a declared transfer map.

Definition 2.4 ([D]). A *compatible realization* is a realization whose certified observable content agrees with the required transfer and persistence conditions.

Definition 2.5 ([D]). *Observational drift* is the certified change in an observable under admissible transfer, prior to identifying whether that change is lawful, bounded, or obstructive.

Definition 2.6 ([D]). *Transfer compatibility* is the condition that a family of observables behaves coherently under the declared class of transfer maps: the transport-class structure is preserved under composition of transfer maps.

Definition 2.7 ([D]). A *transport obstruction* is a certified failure of persistence, compatibility, or lawful transfer. It records the exact stage and context at which a persistent observable fails to transfer lawfully.

Definition 2.8 ([D]). An *observational-transfer object* is an admissible quotient-visible object equipped with the persistence and transport structure needed for lawful comparison across contexts: a stabilized observational quotient, a family of declared transfer maps, and a defect-ledger encoding observational drift along admissible transfer chains.

Definition 2.9 ([D]). A *composable transfer chain* is a finite or admissibly indexed sequence of transfer maps

$$f_0 : (U_0, \Sigma_\infty^{(0)}) \to (U_1, \Sigma_\infty^{(1)}), \quad f_1 : (U_1, \Sigma_\infty^{(1)}) \to (U_2, \Sigma_\infty^{(2)}), \quad \dots$$

for which each successive transfer is lawful and the composite $f_n \circ \cdots \circ f_0$ is well-defined on the relevant certified observable content.

Definition 2.10 ([D]). *Transfer coercivity* is a certified constraint limiting uncontrolled observational drift along admissible transfer chains. It asserts that the accumulated change in a declared observable family across a composable transfer chain is bounded by a quantity determined by the declared structural data.

Remark 2.11 ([R]). Transfer coercivity is the bridge concept that will allow Book IV to assemble local fiber data into a global persistent source object. The coercive bound is what prevents the persistence diagram from diverging as one takes larger and larger regimes: it controls how much observational content can be "lost" at each transfer step.

3. Persistence Lemmas

Lemma 3.1 ([U]). *Persistent observables are invariant under replacement by quotient-equivalent representatives.*

Proof. By Standing Rule 1.1 and the definition of persistent observable: a persistent observable's value is determined by the stabilized quotient class, not by the choice of representative within that class. Two quotient-equivalent representatives belong to the same class, so the observable assigns them the same value. □

Lemma 3.2 ([U]). *Transfer maps preserve certified local compatibility whenever their domain and codomain satisfy the declared admissibility conditions.*

Proof. By Definition 2.2, transfer maps are witness-preserving and compatible with quotient structure. Therefore any certified local compatibility relation in the source context — meaning any admissible relation detectable by the licensed test family — is preserved under the transfer map in the target context. □

Lemma 3.3 ([U]). *Transfer compatibility is invariant under lawful refinement of comparison stage.*

Proof. Suppose a family of observables is transfer-compatible at comparison stage n. Under lawful refinement to stage $n+1$, the quotient Σ_{n+1} refines Σ_n, and transfer maps compatible at stage n remain compatible at stage $n+1$ because the finer partition only introduces additional distinctions that are also quotient-visible. Transfer compatibility, being a condition on equivalence classes, is preserved under refinement. □

Lemma 3.4 ([U]). *Observational drift is well-defined on certified observable families under declared transfer maps.*

Proof. Observational drift measures the change $\mathcal{O}(f(\omega)) - \mathcal{O}(\omega)$ under a transfer map f. Since f is well-defined on quotient classes (Definition 2.2) and $\mathcal{O}$ is a function on quotient classes, the drift is well-defined. □

Lemma 3.5 ([U]). *Transport obstructions are quotient-visible and independent of surplus presentation structure.*

Proof. A transport obstruction is defined as a certified failure of persistence, compatibility, or lawful transfer. Each of these conditions is formulated on quotient classes (by Definitions 2.1, 2.6, 2.2). Therefore the presence or absence of an obstruction is determined at the quotient level; presentation-level surplus cannot introduce or remove a genuine transport obstruction. □

Lemma 3.6 ([U]). *Composition of lawful transfer maps is well-defined on compatible realizations.*

PROOF. If f and g are lawful transfer maps with $\mathrm{codomain}(f) = \mathrm{domain}(g)$, then $g \circ f$ is a witness-preserving map compatible with quotient structure by the definition of transfer maps and the composability condition of Definition 2.9. On compatible realizations, the composition preserves the relevant certified observable content at each step. □

4. Persistence and Compatibility Propositions

PROPOSITION 4.1 ([U]). *Persistent observables are well-defined on the admissible comparison arena.*

PROOF. By Lemma 3.1, persistent observables are quotient-visible. By the Observational Quotient Theorem (Book I, Theorem I.6.1), the quotient Σ_∞ is well-defined. Therefore persistent observables are well-defined functions on Σ_∞. □

PROPOSITION 4.2 ([U]). *Transfer compatibility is preserved under lawful refinement.*

PROOF. Lemma 3.3. □

PROPOSITION 4.3 ([U]). *Compatible realizations induce the same transport class when related by lawful transfer.*

PROOF. If two realizations ω and ω' are related by a lawful transfer map f — meaning $f(\omega) = \omega'$ as quotient-class data — then they are in the same transport class by Definition 2.3. Transfer compatibility ensures this relation is well-defined and extends to the full observable family. □

PROPOSITION 4.4 ([U]). *A transport obstruction prevents identification of the affected observable family as persistent.*

PROOF. A transport obstruction records a certified failure of persistence, compatibility, or lawful transfer (Definition 2.7). A persistent observable must be preserved or lawfully transformed under all declared transfer maps (Definition 2.1). An obstruction certifies that at least one transfer map fails this requirement; hence the affected observable family is not persistent in the declared context. □

PROPOSITION 4.5 ([U]). *Transfer-compatible observable families define a lawful comparison structure across admissible contexts.*

PROOF. Composability of lawful transfer maps (Lemma 3.6) and transfer compatibility (Definition 2.6) together imply that the collection of transfer-compatible observable families, equipped with transfer maps, forms a directed comparison structure: identity transfers are lawful, transfer maps compose, and the transport-class structure is well-defined. □

COROLLARY 4.6 ([U]). *Only transfer-compatible observable families may contribute to the source completion of Book IV.*

PROOF. The source completion in Book IV will assemble fiber data from all admissible regimes into a universal object. Assembly requires that data from different

regimes can be coherently compared, which requires transfer compatibility. Observable families with transport obstructions cannot be coherently assembled, hence they are excluded from the source-completion program. □

5. The Interface-Relative Stability Functional

We now introduce the principal quantitative object of Book III. This functional organizes the balance between bulk stability and boundary burden in a form that is branch-agnostic: it applies to any admissible regime before specialization to any particular endpoint program.

Definition 5.1 ([D]). For a region U_n at comparison stage n, the *bulk stability measure* $B_n^{\mathrm{bulk}}(U_n)$ is the certified coercive content of U_n that is controlled by the Phase-A defect-floor architecture: the net certified observable content on U_n that resists further defect accumulation.

Definition 5.2 ([D]). The *interface burden* $I_n(U_n)$ at stage n is the defect contribution at the boundary $\partial_{\rho^*} U_n$, determined by the LS-2 boundary law: the total certified defect load arising from the collar structure of U_n.

Definition 5.3 ([D]). The *interface-relative stability functional* at stage n is

$$C_n(U_n) \;:=\; B_n^{\mathrm{bulk}}(U_n) \;-\; I_n(U_n).$$

Remark 5.4 ([R]). The interface-relative stability functional C_n is defined at the pre-branch layer, before any specialization to Yang–Mills, Navier–Stokes, gravity, or any other specific endpoint. Its two components are both already present in the certified upstream machinery: B_n^{bulk} from the defect-floor structure of Book I, and I_n from the LS-2 boundary law of Book II. This universality is the source of the functional's load-bearing character.

Definition 5.5 ([D]). The *transport factor* at stage n, denoted $\Theta_n(U_n) \in [0,1]$, is the certified multiplier expressing what fraction of $C_n(U_n)$ survives transport from U_n to U_{n+1} under the admissible enlargement.

Definition 5.6 ([D]). The *transported internal defect* $E_n(U_n)$ is the certified defect increment arising from interior sites of U_n that survive enlargement but are not fully transported: the ledger increments $\Delta\Lambda$ at interior sites during the step $U_n \to U_{n+1}$.

Definition 5.7 ([D]). The *interface/boundary defect* $B_n(U_n)$ is the certified defect increment arising at ∂U_n forced by the LS-2 boundary-determinacy structure: collar-type proliferation at the interface of U_n during the step $U_n \to U_{n+1}$.

Remark 5.8 ([R]). Definitions 5.5–5.7 identify the three sources of observable change across one admissible enlargement step: transported bulk content (discounted by Θ_n), internal defect (E_n), and boundary defect (B_n). The Unified Coercive-Transport Inequality bounds C_{n+1} in terms of these three sources. No other sources are present: the defect ledger of Book I and the boundary law of Book II account for all of them.

6. Finite Balance and the Unified Coercive-Transport Inequality

DEFINITION 6.1 ([D]). A *finite balance datum* is the certified accounting relation describing the transport, gain, loss, or redistribution of observable content across one admissible enlargement step. For the interface-relative stability functional, it takes the form

$$C_{n+1}(U_{n+1}) \;\le\; \Theta_n(U_n)\, C_n(U_n) + E_n(U_n) + B_n(U_n).$$

THEOREM 6.2 ([U]). (Unified Coercive-Transport Inequality.) *For every admissible enlargement $U_n \subseteq U_{n+1}$, the interface-relative stability functional satisfies*

$$C_{n+1}(U_{n+1}) \;\le\; \Theta_n(U_n)\, C_n(U_n) \;+\; E_n(U_n) \;+\; B_n(U_n),$$

where $\Theta_n(U_n) \in [0,1]$ is the transport factor, $E_n(U_n)$ is the transported internal defect, and $B_n(U_n)$ is the interface boundary defect.

PROOF. The bulk stability $B^{\mathrm{bulk}}_{n+1}(U_{n+1})$ of the enlarged regime receives contributions from three distinct sources, which we track using the defect ledger of Book I and the LS-2 boundary law of Book II.

Source 1: Transported bulk. The bulk stability that survives transport from U_n to U_{n+1} is discounted by the transport factor $\Theta_n \in [0,1]$, giving a contribution of at most $\Theta_n C_n(U_n)$. The transport factor accounts for the fact that not all of C_n survives enlargement: some certified content is absorbed by the expansion of the comparison regime.

Source 2: Internal defect E_n. Certified events in the interior of U_n that are not fully transported contribute defect ledger increments $\Delta\Lambda$ at interior sites during the enlargement step. These sum to $E_n(U_n)$, which is additive by the Ledger Additivity Proposition (Book I, Proposition I.5.4).

Source 3: Boundary defect B_n. Certified events at ∂U_n arise from the LS-2 boundary-determinacy structure (Book II, Theorem II.4.1): collar-type proliferation at the interface forces defect contributions $B_n(U_n) = \Delta_{\mathrm{bdry}}(U_n)$. These are determined by the boundary law map Φ_{U_n}.

By additivity of the defect ledger across all three sources,

$$C_{n+1}(U_{n+1}) \;\le\; \Theta_n C_n(U_n) + E_n(U_n) + B_n(U_n). \qquad \square$$

REMARK 6.3 ([R]). The Unified Coercive-Transport Inequality is proved here in fully branch-agnostic form, depending only on the defect ledger of Book I and the LS-2 boundary law of Book II. It is a source-level theorem, upstream of all specializations. The three branch programs — Yang–Mills, Navier–Stokes, gravity — each specialize this inequality by fixing specific forms of Θ_n, E_n, and B_n appropriate to their endpoint targets. Those specializations belong in derived branch books, not in this book. Any citation of this inequality in a branch book must carry an explicit declaration of which specialization is being applied.

COROLLARY 6.4 ([U]). (Terminal-Observable Projection Theorem.) *The interface-relative stability functional C_n admits exactly two canonically distinct functorial projections to terminal observables:*

(i) ***Spectral projection*** $\pi_{\mathrm{YM}}\colon C_n \to g_\infty := \lambda_1(\Delta_A)$*: the covariant Laplacian spectral gap. Closure condition:* $g_\infty > 0$.

(ii) ***Decay projection*** $\pi_{\mathrm{NS}}\colon C_n \to \sup_n Q_n$*: the supremum of the obstruction quotient. Closure condition:* $\sup_n Q_n < \infty$.

Both projections are functorial with respect to the transition maps $\alpha_{R\to R'}$ *of the POT category (Book IV). The two Millennium Prize problems addressed in the YM and NS branch books are the closure conditions of these two projections applied to the same upstream object* C_n.

Proof. Both projections are defined by specializing the decomposition $C_n = B_{\mathrm{bulk}}(U_n) - I_n$ (Definition 5.3). π_{YM} takes B_{bulk} via the covariant Laplacian eigenvalue and I_n via the gap margin; functoriality follows from the transition map coherence of Theorem 6.5. π_{NS} takes C_n via the covering-time–normalized defect floor and Q_n via the quadratic rescaling; functoriality follows from Phase-A exhaustion enlargement compatibility (Theorem 6.6). The projections are distinct because their target observables are dimensionally incomparable: g_∞ is a spectral quantity (units of energy) while $\sup_n Q_n$ is a dimensionless ratio. □

Theorem 6.5 ([U]). *Finite balance data compose along composable transfer chains: if* $U_n \to U_{n+1} \to \cdots \to U_m$ *is a composable transfer chain, then the accumulated interface-relative stability satisfies*

$$C_m(U_m) \;\le\; \Big(\prod_{k=n}^{m-1}\Theta_k\Big)C_n(U_n) \;+\; \sum_{k=n}^{m-1}\Big(\prod_{j=k+1}^{m-1}\Theta_j\Big)\big(E_k(U_k)+B_k(U_k)\big).$$

Proof. Induction on the chain length. The base case $m = n+1$ is Theorem 6.2. For the inductive step, apply Theorem 6.2 to $U_{m-1} \to U_m$ and substitute the induction hypothesis for $C_{m-1}(U_{m-1})$. The transport factors and defect terms compose multiplicatively and additively, respectively, by the Ledger Additivity proposition (Book I). □

Theorem 6.6 ([U]). (Depth-Sum Convergence.) *Let* $(U_n)_{n\ge1}$ *be a Van Hove exhaustion of the canonical substrate: finite connected regions with* $|U_n| \to \infty$ *and* $|\partial U_n|/|U_n| \to 0$. *For any decay rate* $\mu > 0$,

$$\frac{1}{|U_n|}\sum_{d\ge0}|\partial_d U_n|\,e^{-\mu d} \;\longrightarrow\; 0 \quad \text{as } n\to\infty,$$

where $\partial_d U_n$ *denotes the set of sites at distance exactly* d *from the boundary* ∂U_n.

Proof. For any $d \ge 0$, $|\partial_d U_n|$ is bounded above by $|\partial U_n|\cdot D^d$ where D is the maximum degree of the substrate graph. Therefore

$$\sum_{d\ge0}|\partial_d U_n|\,e^{-\mu d} \;\le\; |\partial U_n|\sum_{d\ge0}D^d e^{-\mu d} \;=\; \frac{|\partial U_n|}{1-De^{-\mu}},$$

which is finite for $\mu > \log D$. For general $\mu > 0$, split the sum at depth $d_0 = \lfloor \log(|U_n|)/\mu \rfloor$: the tail $d > d_0$ contributes $|\partial U_n|\cdot O(|U_n|^{-1})$, and the head $d \le d_0$

contributes $|\partial U_n| \cdot d_0$, which is $o(|U_n|)$ by the Van Hove condition $|\partial U_n| = o(|U_n|)$. Dividing by $|U_n|$ gives the result. □

Remark 6.7 ([R]). Theorem 6.6 is branch-agnostic: it uses only the Van Hove exhaustion geometry and is independent of any specific boundary-defect model. It provides the convergence input for the NS Branch Book's weighted-boundary transport bounds, and is placed here rather than in any branch book because it is a structural fact about admissible exhaustions in general.

7. Transfer Coercivity

Definition 7.1 ([D]). An admissible enlargement step $U_n \to U_{n+1}$ is *coercive* if $\Theta_n < 1$, meaning the transport factor is strictly contractive. A composable transfer chain is *uniformly coercive* if there exists $\theta < 1$ such that $\Theta_k \leq \theta$ for all steps k in the chain.

Definition 7.2 ([D]). The defect terms $E_n(U_n)$ and $B_n(U_n)$ are *comparably subordinate* to $C_n(U_n)$ on a regime if there exist branch constants $\mu_E, \mu_B \geq 0$ with $\mu_E + \mu_B < 1 - \theta$ such that $E_n \leq \mu_E C_n$ and $B_n \leq \mu_B C_n$ hold throughout the regime.

Theorem 7.3 ([C]). (Finite Transfer Coercive Inequality.) *Under the declared additional coercivity hypotheses — uniform coercivity with factor $\theta < 1$, and comparably subordinate defect terms — the interface-relative stability functional satisfies*

$$C_m(U_m) \;\leq\; \lambda^{m-n}\, C_n(U_n), \qquad \lambda \;:=\; \theta + \mu_E + \mu_B \;<\; 1,$$

along every admissible composable transfer chain of length $m - n$. In particular, observational drift along admissible transfer chains is quantitatively controlled.

Proof. Under defect subordination, $E_k + B_k \leq (\mu_E + \mu_B)C_k$ at each step. Substituting into the composable balance formula of Theorem 6.5 and bounding the product $\prod \Theta_k \leq \theta^{m-n}$ by uniform coercivity, one obtains

$$C_m(U_m) \;\leq\; (\theta + \mu_E + \mu_B)^{m-n}\, C_n(U_n) \;=\; \lambda^{m-n}\, C_n(U_n).$$

Since $\lambda < 1$, drift is geometrically controlled. □

Remark 7.4 ([R]). Theorem 7.3 is marked conditional because the coercivity hypotheses — $\theta < 1$ and the defect subordination bounds — are extra regime-level burdens not implied by the branch-agnostic upstream machinery alone. At this level of the spine, transport coercivity is part of the source-preparation discipline. Whether it holds for a specific branch requires a branch-level verification. The Navier–Stokes branch, for example, must establish its own defect subordination before citing this theorem.

Corollary 7.5 ([U]). *Observable families carrying transport obstruction are excluded from source-ready observational-transfer structure until the obstruction is resolved.*

Proof. By Proposition 4.4, a transport obstruction certifies that an observable fails to be persistent. By Corollary 4.6, only transfer-compatible persistent observable families may contribute to the source completion. Therefore any family with an unresolved transport obstruction is excluded. □

8. Observational-Transfer Objects and POT Preparation

Proposition 8.1 ([U]). *Observational-transfer objects are well-defined.*

Proof. An observational-transfer object (Definition 2.8) consists of: a stabilized observational quotient (well-defined by Book I, Theorem I.6.1), a family of transfer maps that are well-defined by Definition 2.2 and Lemma 3.6, and a defect ledger (well-defined by Book I, Corollary I.6.3). Each component is well-defined; hence the combined structure is well-defined. □

Proposition 8.2 ([U]). *Lawful transfer maps between observational-transfer objects compose and admit identities.*

Proof. Composition is Lemma 3.6. Identity maps are lawful transfer maps: the identity map is witness-preserving and compatible with quotient structure. Associativity of composition holds because map composition is associative. □

Proposition 8.3 ([U]). *Persistent and transfer-compatible observable families determine observational-transfer structure.*

Proof. Given a persistent observable family (satisfying Definition 2.1) and a transfer-compatible family of transfer maps (satisfying Definition 2.6), the collection of admissible regimes, together with the quotient data and defect-ledger data, constitutes an observational-transfer object by Definition 2.8. □

Proposition 8.4 ([U]). *The collection of observational-transfer objects and lawful transfer maps forms the comparison arena required for source completion.*

Proof. By Propositions 8.1 and 8.2, observational-transfer objects with lawful transfer maps form a well-defined category. By Proposition 8.3, every persistent transfer-compatible observable family yields an object in this category. This is precisely the comparison arena that Book IV's POT construction will use. □

Remark 8.5 ([R]). The observational-transfer arena defined here is the immediate precursor to the POT (Persistent Observational-Transfer) category of Book IV. POT augments these objects with: a representation layer (initially empty, populated only by explicit bridge functors), the full Grothendieck fibration structure over the category of admissible regimes, and the explicit transport invariant $I(R, U)$. All of this additional structure in Book IV builds on the foundation laid here.

Remark 8.6 ([R]). (Run certificates as POT fiber data.) *(Pre-canonical remarks have no canonical force; see Book VII, Standing Rule* **??***.)* The run-certificate structure of the pre-canonical record (v7.35 §2489: $r = (\Sigma, S, T)$ with collar signature Σ, supported type set S, and type operator $T = T_\partial|_S$) is the concrete instantiation of the abstract POT fiber slot defined in Book IV. Two run certificates are comparable

(fiber-equivalent) iff $\Sigma_1 = \Sigma_2$ and $S_1 = S_2$, matching the fiber-over-regime structure of the POT category. This identifies $T_\partial|_S$ as the concrete object populating each fiber; when Book IV is revised to include explicit type-operator language, run certificates will serve as the canonical fiber witnesses.

9. Governing Theorem of Book III

THEOREM 9.1 ([U]). (Observational Transfer Theorem of the Spine.) *Stabilized observables admit lawful transfer across admissible refinements and contexts, yielding the transport structure required for source completion.*

More precisely:

(a) *Persistent observables are well-defined on the admissible comparison arena, are quotient-visible, and are preserved under replacement by quotient-equivalent representatives.*
(b) *Transfer-compatible observable families define a lawful comparison structure across admissible contexts, with transport classes, finite balance data, and well-defined transport obstructions.*
(c) *The interface-relative stability functional C_n satisfies the Unified Coercive-Transport Inequality at every admissible enlargement step, with the three-source decomposition into transported bulk, internal defect, and boundary defect.*
(d) *The collection of observational-transfer objects and lawful transfer maps forms a well-defined comparison arena that serves as the immediate precursor to the POT category of Book IV.*

PROOF. Part (a): Propositions 4.1 and Lemma 3.1.
Part (b): Propositions 4.2, 4.3, 4.5, and Theorem 6.5.
Part (c): Theorem 6.2.
Part (d): Proposition 8.4. □

SCHOLIUM 9.2 ([O]). (Interface-Coherent Directed Compression Schema.) [Proof sketch — retired-note status. Canonical proof open.]

The ICDC Schema is the directed-system formalization of what the Unified Coercive-Transport Inequality means at the scale-limit level: when does a branch family over the enlargement ladder admit a controlled scale-limit compression in which load-bearing branch data survive coherently?

Abstract statement. Given an admissible enlargement ladder $U_{n_0} \subseteq U_{n_0+1} \subseteq \cdots$ from a common source object $\mathcal{U}$, and a branch family (B_n), if the following five conditions hold:

(i) *One-step inheritance:* B_{n+1} inherits load-bearing structure from B_n at each admissible enlargement step.
(ii) *Interface localization of failure:* any failure of inheritance is detectable at the interface ∂U_n.
(iii) *Two-step transitivity of corrections:* interface corrections at step n are absorbed by step $n+2$.

(iv) *Source-image discipline:* the branch data do not supplement the source image with unlicensed external content.

(v) *Summable interface residues:* the accumulated interface defects $\sum_n I_n$ are summable (or suitably subordinate).

then the family admits a controlled scale-limit compression $B_\infty := \varinjlim_\partial B_n$ in which load-bearing branch data survive coherently. The underlying coercive object is $C_n(U_n) = B_n^{\mathrm{bulk}}(U_n) - I_n(U_n)$, satisfying the Unified Coercive-Transport Inequality.

Terminal specializations. The four terminal specializations — YM-ICDC ($\Theta_n = 1$, gap-subcritical), NS-ICDC ($\Theta_n = \theta < 1$, obstruction quasi-decay), G-ICDC (curvature-subcritical), and SCC-ICDC (scale-limit BP-SCC direct system) — belong in their respective branch books as instances of this schema.

Status note. This scholium records an architectural abstraction from the retired dream paper suite (v7.35/v38, ICDC Theorem 9.12). Its five conditions and three-source decomposition are consistent with the canonical UCTI of this book. The canonical proof of the abstract schema remains open and will be hardened when the branch books are written.

Scholium 9.3 ([R]). (Shared Spectral Hinge.)

The spectral gap $m^2 > 0$ derived from the Unified Coercive-Transport Inequality has two distinct terminal readings in the derived branch books:

(i) *Yang–Mills reading (Kato-Rellich route).* In the YM Branch Book, $m^2 > 0$ provides the input to the Kato–Rellich perturbation argument that establishes the Yang–Mills spectral gap $\Delta_{\mathrm{YM}} > 0$ on the Phase-A sector.

(ii) *Navier–Stokes reading (Lieb–Robinson route).* In the NS Branch Book, $m^2 > 0$ determines the Lieb–Robinson decay constant $\mu = m^2\beta > 0$ for the fluid sub-algebra, where $\beta = \log_2(3/2)$ is the canonical transport normalization coefficient (Book IV).

Both readings draw from the same upstream source law and are logically independent of each other. The shared upstream spectral gap is what makes the YM and NS programs structurally related at the level of this book, even though their terminal proofs are entirely different.

Scholium 9.4 ([C]). (Discrete-Time Topological Conservation.) [Conditional on gate structure G1–G3.]

Let $\Phi_t : \mathcal{K}^{(0)} \to S^3$ be a sequence of kernel states. Suppose each time step $t \to t+1$ admits a continuous relative homotopy $H(\cdot, s) : |\mathcal{K}| \to S^3$, $s \in [0,1]$, with $H(\cdot, 0) = \widetilde{\Phi}_t$, $H(\cdot, 1) = \widetilde{\Phi}_{t+1}$, boundary data fixed at all s, and no target-simplex degeneracy along the interpolation (gate G3 absent). Then

$$Q_{\mathrm{simp}}(\Phi_{t+1}) \;=\; Q_{\mathrm{simp}}(\Phi_t).$$

Charge transitions are possible only at declared collapse gates G1–G3. Under strong anchoring ($\Phi_t|_\Gamma = \Phi_\Gamma$ for all t), the conserved charge is the simplicial Hopf invariant.

Source: v7.35 Theorem 66.2. The admissible time evolution of the NFC kernel preserves simplicial topological charge under the collapse gate discipline; this is a structural consequence of the gate structure, not an additional hypothesis.

Scholium 9.5 ([C]). (Hopf Helicity Conservation.) [Conditional on admissibility of Φ.]

Let $\Phi(\cdot, t) : \mathbb{R}^3 \to S^3 \subset \mathbb{C}^2$ be a continuous family of admissible kernel fields with $|\Phi(x,t)| = 1$ for all (x,t), fixed boundary value Φ_∞ (independent of t), and continuous t-dependence. Then the Hopf charge

$$\mathcal{H}[\Phi(\cdot, t)] \;=\; \text{const} \quad \text{in } t.$$

Source: v7.35 Theorem 66.4. The Hopf helicity is an invariant of admissible kernel field evolution; this bridges the discrete NFC collapse gate structure to a classical topological invariant.

Together with Scholium 9.4, this establishes that the NFC kernel carries two conserved topological charges (simplicial charge Q_{simp} and Hopf charge $\mathcal{H}$) under admissible evolution.

10. Boundary of Book III

What Book III proves.

(1) Persistent observables are well-defined, quotient-visible, and transfer-compatible.
(2) Admissible transfer maps compose, are witness-preserving, and respect the observational quotient.
(3) The interface-relative stability functional satisfies the Unified Coercive-Transport Inequality at every admissible enlargement step.
(4) Finite balance data compose along composable transfer chains.
(5) Under extra coercivity hypotheses, observational drift is geometrically controlled.
(6) Observational-transfer objects form a well-defined comparison arena.
(7) Depth-sum convergence for Van Hove exhaustions (Thm 6.6): weighted boundary contributions vanish in the thermodynamic limit.
(8) The ICDC Schema is recorded as an architectural scholium (Sch. 9.2): the abstract UCTI scale-limit formalization.
(9) Two topological conservation scholia (Schs. 9.4–9.5): discrete-time Q_{simp} conservation and Hopf helicity conservation under admissible kernel evolution.

What Book III does not prove, and may not be assumed here.

(1) *Universal source structure.* The existence of a universal source object $U \in$ POT is a theorem of Book IV. Book III builds the transport machinery that makes source completion meaningful; it does not prove source completion itself.
(2) *Branch specializations.* The Yang–Mills specialization ($\Theta_n = 1$, gap-subcritical closure), the Navier–Stokes specialization ($\Theta_n = \theta < 1$, quasi-decay), and the gravity specialization all belong in derived branch books. No specialization of the Unified Coercive-Transport Inequality to a specific endpoint is canonical in this book.
(3) *Continuum structure.* No continuum, differential, or smooth language is licensed at this stage. The transport inequality is a finite discrete bound.

(4) *Branch closure.* Transport coercivity at this level is part of source-preparation discipline. Whether it extends to closure of any specific branch requires branch-level verification.
(5) *Physical endpoint identification.* No identification with gauge fields, fluid dynamics, or geometric structure is claimed.

11. Handoff to Book IV

Book IV begins when the question shifts from *how local certified structure transfers* to *whether all such transfer data can be assembled into a single universal object.* The observational-transfer arena of Book III — objects with stabilized quotients, lawful transfer maps, and defect ledgers — is precisely the POT fiber data that Book IV will assemble into the universal completion.

The transition from Book III to Book IV is:

$$\begin{aligned}&\text{local admissible observables} \longrightarrow \text{stabilized persistent structure} \longrightarrow \\ &\text{lawful transfer across contexts} \longrightarrow \text{universal source completion.}\end{aligned}$$

Book III supplies the first three terms. Book IV supplies the fourth.

12. Dependency Ledger

Theorem	St.	Depends on	Exports to
3.1 Persist. obs. quotient-vis.	[U]	SR 1.1, Def. 2.1	4.1, 9.1
3.2 Transfer preserves compat.	[U]	Def. 2.2	4.2
3.6 Composition well-def.	[U]	Def. 2.9	8.2, 8.4
3.5 Obstructions quot.-vis.	[U]	Defs. 2.1–2.7	4.4, 7.5
4.5 Compat. comparison	[U]	3.6, Def. 2.6	4.6, 9.1
4.6 Source-ready compat.	[U]	4.5	Book IV POT construction
6.2 Unified Coercive-Transport Ineq.	[U]	Books I–II, Defs. 5.3–5.7	6.5, 9.1; all branch books
6.5 Composable balance	[U]	6.2, Bk. I ledger additivity	7.3, 9.1
7.3 Finite Transfer Coercive Ineq.	[C]	6.5, coercivity hypotheses	Branch books (NS, YM, GR imports)
7.5 Obstruction blocks source	[U]	4.4, 4.6	Book IV; branch books
8.4 OT comparison arena	[U]	8.1, 8.2	Book IV POT; 9.1
9.1 Obs. Transfer Thm. of Spine	[U]	All of Bk. III	Books IV–VII; all branch books
6.6 Depth-Sum Convergence	[U]	Van Hove geometry; Bk. I finiteness	NS Branch Book (weighted boundary bounds)
9.2 ICDC Schema (abstract)	[O]	6.2; five conditions	Branch books Part III; YM/NS/GR/SCC ICDC
9.4 Topological conservation	[C]	Gate structure G1–G3	GR Branch Book remark
9.5 Hopf helicity	[C]	Admissibility of Φ	GR Branch Book remark

Scholium 12.1 ([R]). The Unified Coercive-Transport Inequality (Theorem 6.2) is the load-bearing core of Book III and is unconditional: it depends only on the defect

ledger of Book I and the LS-2 boundary law of Book II. The Finite Transfer Coercive Inequality (Theorem 7.3) is conditional, requiring extra coercivity hypotheses. This conditionality should remain explicit in every downstream citation. No branch book may promote the conditional result to unconditional force without discharging the extra hypotheses within the branch.